

Introduction

In the recent project centralising scheduling algorithms, I employed the services of Generative AI (GenAI) tools, specifically ChatGPT, powered by OpenAI's GPT-4 architecture. This AI-driven tool is designed to understand and generate human-like text based on the input it receives. The capabilities of ChatGPT extend beyond simple text generation. From analysing code, offering algorithmic solutions, assisting in debugging, to providing detailed explanations of complex concepts.

Integration ChatGPT into workflow

The integration of GenAI into my workflow was seamless and multifaceted. At the initial stages of my project, I used ChatGPT to clarify topics that I found unclear and get detailed explanations of certain scheduling algorithms such as how lottery scheduling actually works. These explanations helped me understand the theoretical underpinnings of these algorithms before I began coding. During the implementation phase, I consulted ChatGPT to debug my code, optimise algorithms, and format outputs, especially when I encountered issues namely incorrect average time calculations or when my code was running an infinite loop.

Quality of the response and its impact on the solution

Although the genAI sometimes can overlook potential problems arising from changes when providing suggestions, the overall quality and correctness of generated texts from ChatGPT was personally consistent. I was able to properly format output Strings, learning about how `String.format()` can help me achieve the desired output. Moreover, chatGPT was especially helpful when it comes to optimising and shortening my code. It analyses long paragraphs of codes, providing an example of how it could be shortened if it could be written more elegantly, resulting in neat, optimised code, removing unnecessary parts. From time to time, it also correctly detects faulty logic that I overlooked, answering the question why my code was not running as expected.

Improvement in Learning Experience and Problem-Solving Skills

Using GenAI tools has created a good impact on problem-solving skills. The ability to ask questions and receive instant, detailed explanations helped solidify my understanding of complex concepts. Even if the provided solution was not determined enough, elaboration deeper in the topics or examples was provided, enhancing my learning experience. This interactive learning process allowed me to bridge gaps in my

knowledge more effectively. Additionally, the GenAI tool also provided different approaches to a certain problem, therefore giving me more point of view on the encountered challenge, leading to the fact that I can freely choose which approach could be more effective, more suitable for the project. I was able to develop a clearer understanding of the overall structure and logic required to implement a solution.

Challenges Faced and Solutions

One of the main challenges I faced while using GenAI tools was the occasional difficulty in interpreting code-related suggestions. Since the AI generates responses based on patterns learned from vast datasets, there were times when the suggestions were not entirely aligned with the specific requirements of my project. Another challenge was ensuring that I remained an active participant in the problem-solving process rather than relying too heavily on the AI. To address this, I made it a point to understand and analyse each suggestion before implementing it, using the tool as a guide rather than a crutch. Lastly, the genAI can misinterpret my code and my intentions from time to time. I learned to frame my queries more precisely and provide context to ensure that the AI understood my requirements fully.

Productivity and Time Saved

The GenAI tool significantly boosted my productivity by reducing the time spent on debugging, researching, and understanding complex concepts. Tasks that could have taken up to hours, such as debugging a code-written scheduling algorithm, finding optimal functions to support the scheduling algorithm, were accomplished in less time with the help of ChatGPT. This time-saving aspect allowed me to focus more on refining my solution and exploring additional features. Moreover, the tool's ability to provide instant feedback and run code snippets in real-time meant that testing ideas and solutions may take place immediately, resulting in quicker iterations and a more efficient development process.

Ethical Considerations

While the use of GenAI tools provided numerous benefits, it also raised certain ethical considerations. One major concern is the potential for over-reliance on AI-generated solutions, which could lead to a diminished understanding of fundamental concepts. To mitigate this, I consciously make sure that seeking suggestions from ChatGPT shall only be the last approach in order to resolve a problem. By fully grasping the reasoning behind each suggestion, I believe this can be seen as a learning opportunity. Additionally, there is the question of data privacy. While using GenAI tools, I made sure

not to input any sensitive or personal information, understanding that the AI processes data based on large, anonymized datasets.

Conclusion

In conclusion, the integration of GenAI tools like ChatGPT into my workflow has been a highly positive experience. The tools have improved my understanding of complex concepts, enhanced my problem-solving skills, and significantly boosted my productivity. However, it is important to use these tools ethically, ensuring that they complement rather than replace traditional learning methods. By striking this balance, I was able to make the most of what GenAI has to offer, leading to a successful and enriching project outcome.